Sec: OSR.IIT_*COSC(MODEL-A&APEX) | GTA-8(P1) | Date: 23-04-26
Time: 3 Hrs | 2025_P1 | Max. Marks: 180

KEY SHEET

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MATHEMATICS

1	B	2	C	3	D	4	B	5	BC
6	AB	7	AD	8	3	9	6	10	2.50
11	148	12	48	13	4	14	A	15	D
16	D								

PHYSICS

17	A	18	C	19	A	20	C	21	ABCD
22	ABC	23	AC	24	6.32 TO 6.33	25	1552.00 TO 1554.00	26	0.04 TO 0.06
27	2.20 TO 2.24	28	40	29	106.66 TO 106.67	30	D	31	A
32	A								

CHEMISTRY

33	B	34	D	35	C	36	A	37	AD
38	AB	39	ABC	40	1.70 TO 1.73	41	8.50 TO 8.55	42	3.20
43	7	44	40.00 TO 40.20	45	37	46	D	47	C
48	B								

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SOLUTIONS
MATHEMATICS

1. $x^3-6x^2+12x-1=0$ or $x^3-6x^2+12x-8=-7$

or $(x-2)^3=-7$

or $x-2=-7^{1/3},-7^{1/3}\omega,-7^{1/3}\omega^2$

or $x=2-7^{1/3},2-7^{1/3}\omega,2-7^{1/3}\omega^2$

$\Rightarrow$ only one irrational root.

2. P (at least 2 of them score above 75% marks)

$=\frac{1}{2}-\frac{1}{4}=\frac{1}{4}$ $P(E_3)=\frac{P(S_3 \text{ and } S_1 \text{ or } S_3 \& S_2 \text{ score above} 751\% \text{ marks})}{P(\text{ at least} 2 \text{ of them score above} 751\% \text{ marks})}=\frac{1}{10}\Rightarrow P(S_3$ and S_1 or S_3 & S_2 score above 75% marks)

$=\frac{1}{40}$

P(only S_1 & S_2 score above 75%)

$=\frac{1}{4}-\frac{1}{40}=\frac{9}{40}$ P (only S_1 or S_2 and not S_3 score above 75%)

$=\frac{1}{2}-\frac{1}{5}-\frac{9}{40}=\frac{3}{10}-\frac{9}{40}=\frac{3}{40}$ P (only S_3 scores above 75%)

$=\frac{1}{4}-\left(\frac{3}{40}\right)=\frac{7}{40}$

3. Observe that L & P are perpendicular implying P_1 is perpendicular to P . Also observe that Q lies in plane P .

$2(2\lambda+1)+9\lambda+\lambda+2-18=0\ 14\lambda+4-18=0\ 14\lambda=14\Rightarrow\lambda=1$ $P(3,3,3)$ & $Q(5,4,-4)$ PQ is normal vector to the plane.

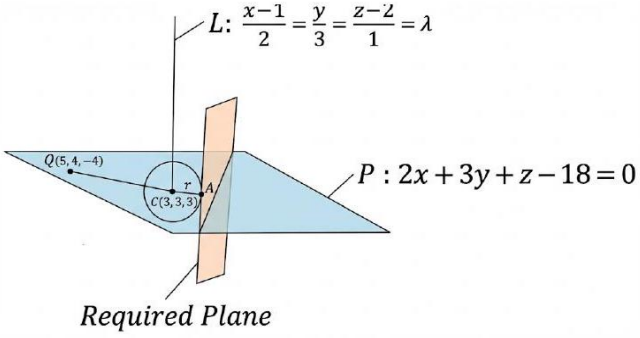
$\vec{n}=2\hat{i}+\hat{j}-7\hat{k}$ equation of plane is $2x+y-7z+k=0$

distance from $(3,3,3)$ is 2

$\Rightarrow\frac{|k-12|}{\sqrt{54}}=\frac{|k-12|}{3\sqrt{6}}=\sqrt{6}\Rightarrow|k-12|=18\Rightarrow k=-6,30\Rightarrow 2x+y-7z-6=0$ or $2x+y-7z+30=0$

Distance of $2x+y-7z+30=0$

from ($5,4,-4$) is maximum



4. $A^2B=BA$ Post multiplying by A

$A^2(BA)=BA^2$ $A^4B=BA^2$ $A^4BA=BA^3$ $A^6B=BA^3\Rightarrow BA^3=A^6B$ Pre-multiplying by B

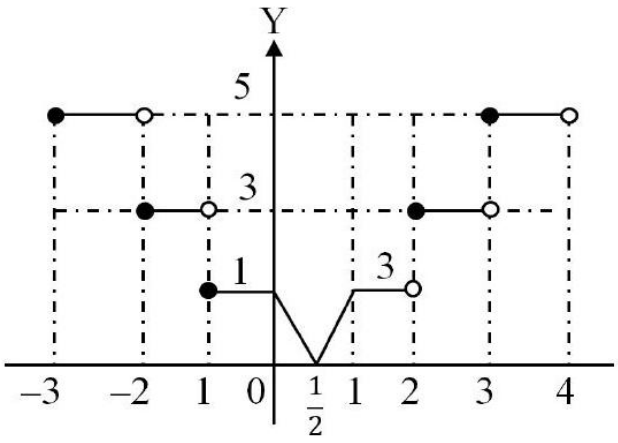
$\Rightarrow B^2A^3=BA^6B=A^2BA^5B=A^4BA^4B=A^6BA^3B=A^8BA^2B=A^{10}BAB=A^{12}B^2$ Again pre-

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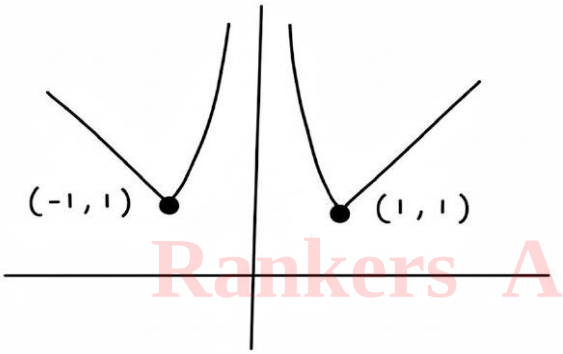
multiplying by B

$\Rightarrow B^3\ A^3 = BA^{12}\ B^2 = BA^6\ A^6\ B^2 = A^{12}BA^6\ B^2 = A^{24}\ B^3 \Rightarrow P^3\ B^3\ A^3 = P^3\ A^{24}\ B^3 = A^{21}\ B^3$

5. $f(x) = \left|\left|[x]+\{x\}\right|+\left|\{x\}-[x]\right|-1\right| = \left|2\max\left(\left|[x]\right|,\{x\}\right)-1\right|$



$g(x) = x, x \geq 1 = \frac{1}{x}, x \in (0,1) = -\frac{1}{x}, x \in (-1,0) = -x, x \leqslant -1$



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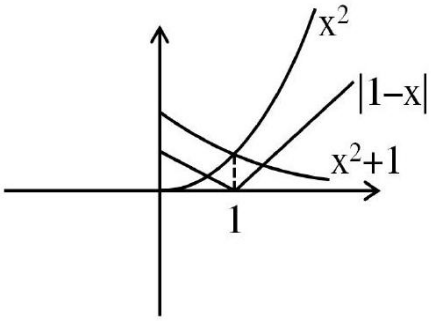
6. $f(x).f(y)-f(x)-f(y)+1 = f(xy)-1 \Rightarrow (f(x)-1)(f(y)-1) = f(xy)-1$ let $f(x)-1 = g(x)$

$\Rightarrow g(x)\cdot g(y) = g(xy) \Rightarrow g(x)$ can be $x^n, 0$

$\Rightarrow f(x)$ can be $x^n+1, 1$

but $f(x)$ is one-one

$\Rightarrow f(x) = x^n+1\ f'(1) = 2 \Rightarrow n = 2 \Rightarrow f(x) = x^2+1 \Rightarrow f(5) = 26\ h(x) = \min\left\{\frac{2}{x^2+1}, x^2, |1-x|\right\}, x \geq 0$



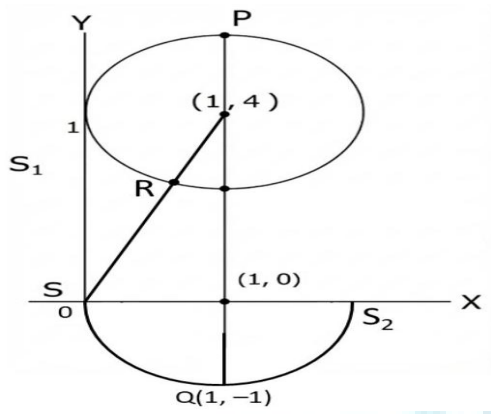
$\Rightarrow \frac{2}{x^2+1} = x^2 \Rightarrow x^4+x^2-2 \Rightarrow x = 1 \because x \geq 0\ x^2 = |1-x| \Rightarrow (x^2)^2 - (x-1)^2 = 0$

$\Rightarrow (x^2+x-1)(x^2-x+1) = 0 \Rightarrow x = \frac{-1+\sqrt{5}}{2} \because x \geq 0$

3 points of non differentiability

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7.



$S_1 : (x-2)^2+(y-3)^2+x^2+(y-5)^2=8$ or $S_1 : (x-1)^2+(y-4)^2=2$

and S_2 is semicircle with centre $(1,0)$ and radius 1 as shown.

maximum distance is $PQ=5+\sqrt{2}$ and minimum distance is $RS=\sqrt{17}-\sqrt{2}$

8. $\vec{c}=\vec{a}\times\left(\vec{a}\times\vec{b}\right)=\pm\left(-\hat{j}+\hat{k}\right)$

$\vec{d}\,\|\left(\vec{a}\times\vec{b}\right)\Rightarrow\vec{d}$ is perpendicular to the plane of $\vec{a}\vec{b}\,\&\,\vec{c}$

$$[\vec{b}\vec{c}\vec{d}]^2=\begin{vmatrix}6 & \pm3 & 0\\ \pm3 & 2 & 0\\ 0 & 0 & 3\end{vmatrix}=36-27=9$$

$$\Rightarrow\left|\left[\vec{b}\vec{c}\vec{d}\right]\right|=3$$

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9. Case-I : When n is odd

⇒ Let n consecutive integers be

$$r-\frac{(n-1)}{2}\ldots r-1,r,r+1\ldots r+\frac{n-1}{2}$$

where r is median

$$\mu=\frac{2\left[(r-r)+(r+1-r)\ldots\left(r+\frac{n-1}{2}-r\right)\right]}{n}$$

$$=\frac{2\left(1+2\ldots\frac{n-1}{2}\right)}{n}$$

$$=\frac{2\left(\frac{n-1}{2}\right)\left(\frac{n+1}{2}\right)}{2n}$$

$$=\frac{n^2-1}{4n}=\frac{n}{4}-\frac{1}{4n}$$

If $\frac{n}{4}-\frac{1}{4n}$ is terminating

$\Rightarrow\frac{1}{4n}$ should be terminating

⇒ So number of values of n for which μ is non terminating is

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$\frac{1000-10}{2}-3=492$

Case-II : When n is even (say 2 k)

$\Rightarrow$ Let n consecutive terms be

$r-k+\frac{1}{2}\ldots r-\frac{3}{2},r+\frac{1}{2},r+\frac{3}{2}\ldots r+k-\frac{1}{2}$

where $\frac{r-\frac{1}{2}+r+\frac{1}{2}}{2}=r$ is median

$\Rightarrow \mu=\frac{2\left[\left(r+\frac{1}{2}-r\right)+\left(r+\frac{3}{2}-r\right)+\ldots+\left(r+k-\frac{1}{2}-r\right)\right]}{2k}$

$=\frac{1+3+\ldots(2k-1)}{2k}=\frac{k^2}{2k}$

$=\frac{k}{2}=\frac{n}{4}$

$\frac{n}{4}$ is always terminating

$\Rightarrow$ number of even values of n for which n is non terminating is 0

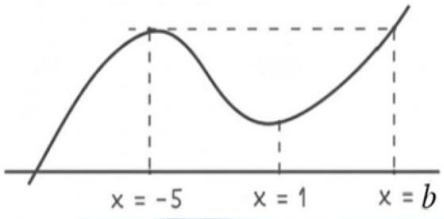
So $\frac{\lambda}{82}=\frac{492+0}{82}=\frac{492}{82}=6$

10. $f'(x)=3x^2+12x-15$

$=3\left(x^2+4x-5\right)$

$=3(x+5)(x-1)$

$\Rightarrow c=-5$ or 1



$f(x)=x^3+6x^2-15x+3$

For maximum possible value of b such that

$f(a)=f(b),a=-5,c=1$

$\Rightarrow -5-5+b=-6$

$\Rightarrow b=4$

$\Rightarrow g(x)=4x^2-5x^2+x+d$

$g'(x)=12x^2-10x+1<0\forall x\in(e,f)$

$\Rightarrow e+f=\frac{5}{6}$

$\Rightarrow 3(e+f)=2.5$

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11. $S: x^2+y^2=16$

$E: \frac{x^2}{64}+\frac{y^2}{4}=1$

For $y=mx+c$ to be common tangent

$\pm 4\sqrt{1+m^2}=\sqrt{64\,m^2+4}$

$16+16\,m^2=64\,m^2+4$

$12=48\,m^2$

$m=-\frac{1}{2}$

$\Rightarrow y=-\frac{1}{2}x\pm 4\sqrt{1+\frac{1}{4}}$

or $y=-\frac{1}{2}x+2\sqrt{5}$ is the common tangent

Since it is normal to the parabola $y^2=4ax$

$\Rightarrow -2am-am^3=2\sqrt{5}$

solving, $a=\frac{16\sqrt{5}}{9}$

Also, $t_1+t_2=-4$ and $t_1+t_2=\frac{-2}{t_1}$

$\Rightarrow t_1=\frac{1}{2}\Rightarrow x_1=\frac{4\sqrt{5}}{9}$ and $y_2=-16\sqrt{5}$

$\Rightarrow x_1+y_2=\frac{\sqrt{5}}{9}\#(148)$

12. $\frac{1}{30}\int\frac{-30\left(\frac{1}{x^6}+\frac{1}{x^7}-\frac{1}{x}\right)dx}{\left(\frac{6}{x^5}+\frac{5}{x^6}+30\log_e x\right)^{7/8}}$

$\frac{1}{30}\int\frac{\left(-30x^{-6}-30x^{-7}+\frac{30}{x}\right)dx}{\left(6x^{-5}+5x^{-6}+30\log_e x\right)^{7/8}}=\frac{1}{30}\left(\frac{t^{-7/8}}{\frac{-7}{8}+1}\right)=8t^{1/8}$

$=\frac{8}{30}\left(6x^{-5}+5x^{-6}+30\log_e x\right)^{1/8}+c$

$\because f(x)=\frac{6}{x^5}+\frac{5}{x^6}+30\ln x$

$\Rightarrow f'(x)=-\frac{30}{x^6}-\frac{30}{x^7}+\frac{30}{x}$

$=\frac{30}{x}\left(1-\left(\frac{1}{x^5}+\frac{1}{x^6}\right)\right)<0\forall x\in\left(\frac{1}{2},1\right)$

and $f(1)=11$

$\Rightarrow f(x)\geq 11\forall x\in\left(\frac{1}{2},1\right)$

$\Rightarrow A=\int_{1/2}^{1}\left(6x^{-5}+5x^{-6}+30\ln x-11\right)dx$

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$$=\left(-\frac{3}{2}x^{-4}-x^{-5}+30x\ln x-30x-11x\right)\Big|_{1/2}^{1}$$

$$=-\frac{3}{2}-1-41-\left(-\frac{3}{2}\cdot 16-32+15\ln 2-15-\frac{11}{12}\right)$$

$$=-\frac{5}{2}-41+24+32-15\ln 2+15+\frac{11}{2}$$

$$=33-15\ln 2$$

$p=33, q=15$

$\Rightarrow p+q=48$

13. $f(x)=\frac{1}{2}\left(\tan^{-1}\left(x-\frac{1}{2}\right)+\{-x\}+\{x\}\right)$

$$f(1-x)=\frac{1}{2}\left(\tan^{-1}\left(\frac{1}{2}-x\right)+\{x\}+\{-x\}\right)$$

$\Rightarrow f(x)+f(1-x)=\{x\}+\{-x\}$

Also,

$$I=\int_{\frac{1}{2025}}^{\frac{2}{2025}}\left(f(x)+f\left(x+\frac{1}{2025}\right)+\ldots+f\left(x+\frac{2022}{2025}\right)\right)dx$$

$$=\int_{\frac{1}{2025}}^{\frac{2024}{2025}}f(x)\,dx$$

$$\Rightarrow \int_{\frac{1}{2025}}^{\frac{2024}{2025}}f(1-x)\,dx$$

$$\Rightarrow 2I=\int_{\frac{1}{2025}}^{\frac{2024}{2025}}1\,dx=\frac{2023}{2025}$$

$$\Rightarrow I=\frac{2023}{4050}\Rightarrow\left\{\frac{4050}{2023}\right\}=\frac{4}{2023}$$

$\Rightarrow x=4$

14. (P) Number of a asymmetric relation $=3^{\frac{n^2-n}{2}}$

$=3^{325}=p^q$

$\Rightarrow p+q=328$

(Q) $n(A\times A)=26\times 26$

For reflexive relation (a, a) must be taken hence rest of the selection will be from $26\times 26-26=650$ elements of $A\times A$

$\Rightarrow$ Total number of reflexive relations are 2^{650}

Total number of unordered pairs (a,b),

$a\neq b$ are ${}^{26}C_2=325$.

For each such pair, include both or exclude both

$\Rightarrow$ 2 choices each

So, reflexive + symmetric relations are 2^{325}

Hence reflexive but not symmetric are

$2^{650}-2^{325}=2^{325}\left(2^{325}-1\right)$ (R) Total number of unordered pairs (a,b), $a\neq b$ are ${}^{26}C_2=325$.

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Total ways of choosing (a,b) & (b,a) simultaneously but not all (a,a) are

$\left({}^{325}C_0+{}^{325}C_1+{}^{325}C_2+\ldots+{}^{325}C_{325}\right)\left({}^{26}C_0+{}^{26}C_1+{}^{26}C_2+\ldots+{}^{26}C_{25}\right)-1=2^{325}\left(2^{26}-1\right)-1$ (S) Total non-empty relations are $2^{676}-1$

Symmetric relation are $2^{325}\cdot 2^{26}=2^{351}$

Reflexive relations are 2^{650}

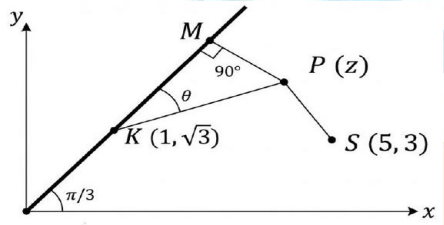
Symmetric and reflexive relations are 2^{325}

$\Rightarrow$ Neither symmetric nor reflexive are

$2^{676}-2^{650}-2^{351}+2^{325}-1+1=2^{325}\left(2^{325}-1\right)\left(2^{26}-1\right)$

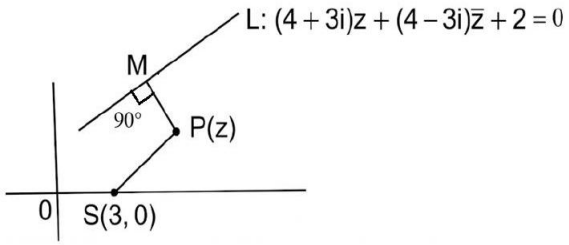
15. (P) $\dfrac{\left|z-(5+3i)\right|}{\left|z-1-i\sqrt{3}\right|}=\sin\left(\dfrac{\pi}{3}-\arg\left(z-1-i\sqrt{3}\right)\right)$

$\theta=\dfrac{\pi}{3}-\arg\left(z-1-i\sqrt{3}\right)\ \dfrac{PS}{PK}=\sin\theta\Rightarrow PS=PM$ locus is parabola



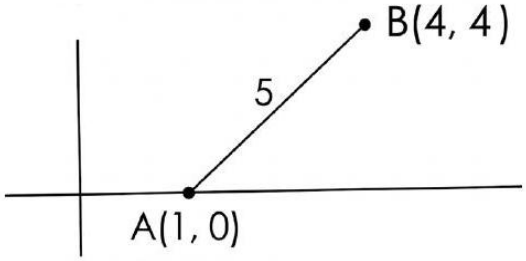
(Q) $|z-3|=\dfrac{2\left|(4+3i)z+(4-3i)\bar{z}+2\right|}{10}$

$|z-z_1|=\dfrac{2|\bar{a}z+\bar{a}z+b|}{2|a|}\Rightarrow PS=2PM\Rightarrow$ locus is hyperbola



(R) $\left|\,|z-1|-|z-4-4i|\,\right|=5$

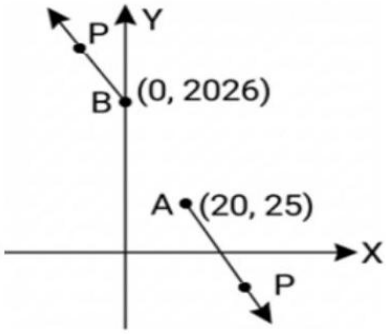
$|PA-PB|=AB\Rightarrow z$ lies on pair of rays



(S) $\arg\left(\dfrac{z-20-25i}{z-2026i}\right)=\pi$

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$\arg(PA)-\arg(PB)=\pi \Rightarrow P$ lies on pair of rays



16. (P) $\frac{{}^{n}C_{r}}{{}^{r+k}C_{r}}=\frac{{}^{n+k}C_{r+k}}{{}^{n+k}C_{k}}$

$\Rightarrow \frac{{}^{25}C_{r}}{{}^{r+10}C_{r}}=\frac{{}^{35}C_{r+10}}{{}^{35}C_{10}} \Rightarrow \sum_{r=0}^{10}\frac{{}^{25}C_{r}\cdot{}^{10}C_{10-r}}{{}^{r+10}C_{r}}=\frac{1}{{}^{35}C_{10}}\sum_{r=0}^{10}{}^{35}C_{r+10}\cdot{}^{10}C_{10-r}=\frac{{}^{45}C_{20}}{{}^{35}C_{10}}=\frac{{}^{45}C_{10}}{{}^{20}C_{10}}$ (Q)

${}^{n}C_{r}\cdot{}^{n-r}C_{k-r}={}^{n}C_{k}\cdot{}^{k}C_{r}$

$\Rightarrow {}^{25}C_{r}\cdot{}^{25-r}C_{10-r}={}^{25}C_{10}\cdot{}^{10}C_{r} \Rightarrow \sum_{r=0}^{10}{}^{15}C_{10-r}\,{}^{25}C_{r}\,{}^{25-r}C_{10-r}={}^{25}C_{10}\sum_{r=0}^{10}{}^{15}C_{10-r}\,{}^{10}C_{r}={}^{25}C_{10}\cdot{}^{25}C_{10}$ (R)

${}^{n}C_{r}\cdot{}^{n-r}C_{k}={}^{n}C_{k}\cdot{}^{n-k}C_{r}$

${}^{25}C_{r}\cdot{}^{25-r}C_{9}={}^{25}C_{9}\cdot{}^{16}C_{r} \Rightarrow \sum_{r=0}^{16}(-1)^{r}\,{}^{25}C_{r}\,{}^{25-r}C_{9}\cdot{}^{16}C_{r}={}^{25}C_{9}\sum_{r=0}^{16}(-1)^{r}\cdot{}^{16}C_{r}\cdot{}^{16}C_{r}=(-1)\cdot{}^{8}\cdot{}^{25}C_{9}\cdot{}^{16}C_{8}$(

S) ${}^{n}C_{r}\cdot{}^{r}C_{k}={}^{n}C_{k}\cdot{}^{n-k}C_{r-k}$

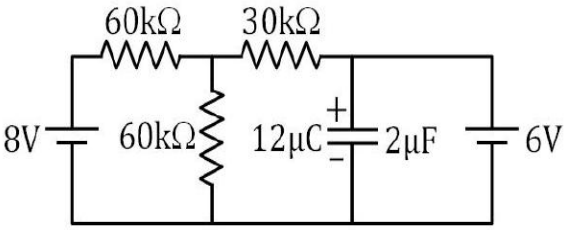
$\Rightarrow {}^{28}C_{r}\cdot{}^{r}C_{10}={}^{28}C_{10}\cdot{}^{18}C_{r-10} \Rightarrow \sum_{r=10}^{18}(-1)\cdot{}^{r}\cdot{}^{28}C_{r}\cdot{}^{r}C_{10}\cdot{}^{18}C_{r}$

$={}^{28}C_{10}\sum_{r=10}^{18}(-1)^{r}\cdot{}^{18}C_{r-10}\,{}^{18}C_{18-r}={}^{28}C_{10}\cdot{}^{18}C_{4}$

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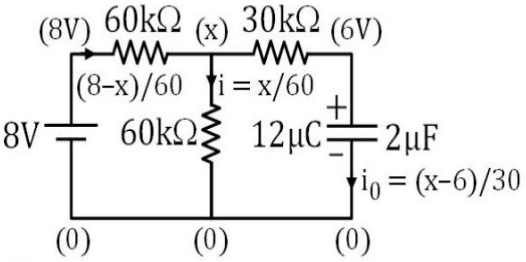
PHYSICS

17. Just before $t=0$



$q_0=12\mu C$

Just after $t=0$



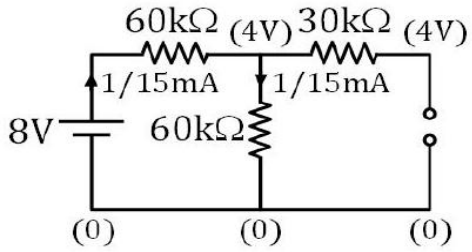
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$$\frac{8-x}{60}=\frac{x}{60}+\frac{x-6}{30}$$

$$8-x=x=2x-12$$

$$x=5\Rightarrow i_0=\frac{-1}{30}\ mA$$

At steady state :-

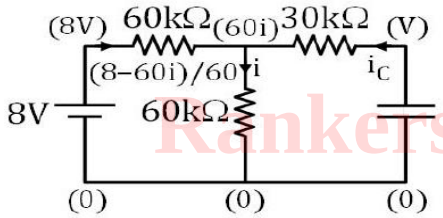


$$q_{ss}=(4-0)\times 2=8\mu C$$

time constant,

$$\tau=\frac{q_{ss}-q_0}{i_0}=\frac{(8-12)\times 10^{-6}}{-\frac{1}{30}\times 10^{-3}}=0.12sec$$

At time t ,



voltage across capacitor as function of time is

$$v(t)=6+(4-6)(1-e^{-t/0.12})=4+2e^{-t/0.12}$$

current through capacitor as function of time is

$$i_C(t)=\frac{t}{30}e^{-t/0.12}\ mA$$

Also, $V-60i=30i_C$

$$\Rightarrow 4+2e^{-t/0.12}-60i=30\times\frac{1}{30}e^{-t/0.12}$$

or, $4+2e^{-t/0.12}=60i$

$$\Rightarrow i=\frac{1}{15}+\frac{1}{60}e^{-t/0.12}\ mA$$

$$=\frac{1000}{15}+\frac{1000}{60}e^{-t/(120\ ms)}\mu A$$

$$=66.67+16.67e^{-t/(120\ ms)}\mu A$$

18. Correct Answer is (C)
19. Correct Answer is (A)

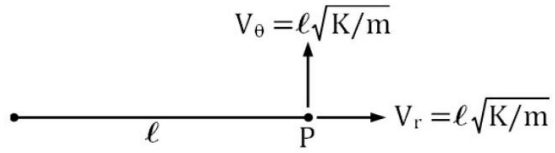
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20. $R_{eq} = R_1 + R_2$

$$= \frac{\left(\frac{1}{R} - \frac{1}{2R}\right)}{4\pi K} + \frac{\left(\frac{1}{2R} - \frac{1}{3R}\right)}{4\pi 2K} = \frac{1}{4\pi KR}\left(\frac{1}{2} + \frac{1}{12}\right) = \frac{7}{48\pi KR}$$

$$H = \frac{9\ T}{7/48\pi KR} = \frac{432}{7}\pi KRT$$

21.



Angular momentum, $L = m\ell^2\sqrt{\frac{K}{m}}$ at $t = 0$

P.E. : $U = \int_0^\ell Krdr = \frac{K\ell^2}{2}$

at max / min distance r_m, speed v_m

$$mv_m r_m = m\ell^2\sqrt{\frac{K}{m}} \Rightarrow v_m r_m = \ell^2\sqrt{\frac{K}{m}} \#(1)$$

and $\frac{1}{2}mv_m^2 + \frac{1}{2}Kr_m^2 = \frac{1}{2}K\ell^2 + \frac{1}{2}m2\ell^2\frac{K}{m}$

or, $\frac{1}{2}m\frac{\ell^4}{r_m^2}\frac{K}{m} + \frac{1}{2}Kr_m^2 = \frac{K\ell^2}{2} + K\ell^2$

$$\frac{\ell^2}{2r_m^2} + \frac{1}{2}\left(\frac{r_m}{\ell}\right)^2 = \frac{3}{2}$$

$$\left(\frac{r_m}{\ell}\right)^4 - 3\left(\frac{r_m}{\ell}\right)^2 + 1 = 0$$

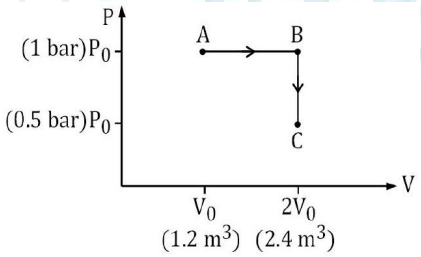
$$\left(\frac{r_m}{\ell}\right)^2 = \frac{+3 \pm \sqrt{9-4}}{2\times 1} = \frac{3 \pm \sqrt{5}}{2}$$

$\Rightarrow r_{max} = 1.62\ell; r_{min} = 0.62\ell$

In radial direction, the force is similar to spring force.

Hence, time period $= 2\pi\sqrt{\frac{m}{K}}$

22.



$$\Delta Q = \Delta W + \Delta U = 10^5 \times 1.2\ J = 120\ kJ$$

$$\Delta S = n\frac{7R}{2}\ell n(2) + n\frac{5R}{2}\ell n\left(\frac{1}{2}\right) = nR\ell n2$$

$$= \frac{10^5 \times 1.2}{300} \times 0.7 = 0.28\ kJ/k$$

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For $AB: dS = n\frac{7R}{2}\frac{dT}{T} \Rightarrow \frac{dT}{dS} = \frac{2\ T}{7nR}$

T increases $\Rightarrow$ slope $\frac{dT}{dS}$ increases

Similarly, for BC, T decreases $\Rightarrow$ slope decreases

23. $\frac{Q\omega}{2\pi}\pi R^2\ B \leqslant mg\frac{L}{2}$

$BQVR \leq mgL$

$v \leqslant \frac{mgL}{BQR}$

24. $Z = \frac{(1+i)(-i)}{(1+i-i)} + 1 = 2 - i$

$I = \frac{10i}{2-i} = \frac{10i(2+i)}{5}$

$= 2(-1+2i) = -2+4i$

$V(t) = 10i - 1\ T = 10i - (-2+4i) = 2 - 6i$

$V_{max} = \sqrt{2^2+6^2} = 2\sqrt{10}\ V = 6.32\ V$

25. $F_V = \eta A\frac{dV}{dr}$

$= 860 \times 0.00042 \times 2\pi rL \times \frac{V}{\Delta r}$

$= 860 \times 0.00042 \times 2\pi \times 0.14 \times 2 \times \frac{0.22}{0.09 \times 10^{-3}}\ N$

$= 1553.34\ N$

26. $8.53 = M_{N^{14}} + M_d - M_P - M_{N^{15}}\ \#(1)$

$7.58 = M_{N^{15}} + M_d - M_\alpha - M_{e^{13}}\ \#(2)$

$5.16 = M_{C^{13}} + M_d - M_\alpha - M_{B^{11}}\ \#(3)$

$Q = M_{B^{11}} + M_\alpha - M_n - M_{N^{14}}\ \#(4)$

$(1)+(2)+(3)+(4)$:

$\Rightarrow 21.27 + Q = 3M_d - M_P - M_n - M_\alpha$

$Q = (6.0420 - 1.0078 - 1.0087 - 4.0026) \times 931 -\ 21.271\ eV$

$= 0.0499MeV = 0.05MeV$

27. When bead is displaced by x , restoring force on it is,

$F_{res} = \frac{GM(M/L)2x}{(L/2)^2} = \frac{8GMm}{L^3}x$

$m\omega^2 x = \frac{8GMm}{L^3}x$

$\omega = \sqrt{\frac{8GM}{L^3}}$

$T = 2\pi\sqrt{\frac{L^3}{8GM}} = \frac{\pi}{\sqrt{2}}\sqrt{\frac{L^3}{GM}} = 2.22$

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28. $5=(2n+1)\frac{\lambda}{4}$

$15=(2\,m+1)\frac{\lambda}{4}$

$80=2\,K\times\frac{\lambda}{4}$

$\Rightarrow \frac{2\,K}{16}=\frac{2\,m+1}{3}=\frac{2n+1}{1}$

$K=16n+8$

$m=n+1$

For minimum frequency, $n=0,\ m=1,\ K=8$

$\Rightarrow \frac{\lambda}{4}=5\text{ cm}$

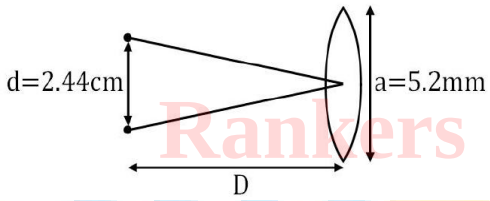
$\Rightarrow \lambda=20\text{ cm}=0.2\text{ m}$

$V=\sqrt{\frac{Y}{\rho}}=\sqrt{\frac{1.6\times10^{11}}{2500}}=\frac{4}{5}\times10^{4}\text{ m/s}$

$=8000\text{ m/s}$

$f=\frac{\nu}{\lambda}=40000\text{ Hz}$

29.



$D\leqslant\frac{da}{1.22\lambda}$ for resolving

$\frac{da}{1.22\lambda_1}\leqslant D\leqslant\frac{da}{1.22\lambda_2}$

$\frac{2.44\times10^{-2}\times5.2\times10^{-3}}{1.22\times650\times10^{-9}}\text{ m}\leqslant D\leqslant\frac{2.44\times10^{-2}\times5.2\times10^{-3}}{1.22\times390\times10^{-9}}\text{ m}$

$1.6\times10^{2}\text{ m}\leqslant D\leqslant\frac{8}{3}\times100\text{ m}$

$160\text{ m}\leqslant D\leqslant266.67\text{ m}$

$\Rightarrow \beta=266.67\text{ m},\alpha=160\text{ m}$

$\beta-\alpha=106.67\text{ m}$

30. (P) $\frac{hc}{\lambda_{min}}=eV_0$

(Q) Change in target metal result in changing characteristic wavelengths.

(R) Frequency of K_α obeys Mosley's law i.e. $f\propto(\tau-1)^2$.

(S) Increasing the current means increasing number of incident electrons which results in increasing intensity with out disturbing wavelengths.

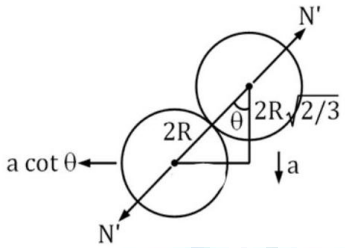
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31. $3\,N\sqrt{\frac{2}{3}} = mg \Rightarrow N = 10\sqrt{\frac{3}{2}}$

$Ng = \frac{4}{3}mg = 40$

$2\,T\cos 30^\circ = N \times \frac{1}{\sqrt{3}} \Rightarrow T = \frac{10}{\sqrt{6}}$

Upon cutting the string



$N'\sin\theta = ma\cot\theta$

or, $N' = 3a\sqrt{6}$

$mg - 3\,N'\cos\theta = ma \#(1)$

or, $10\sqrt{\frac{3}{2}} - N' = \sqrt{\frac{3}{2}}a$

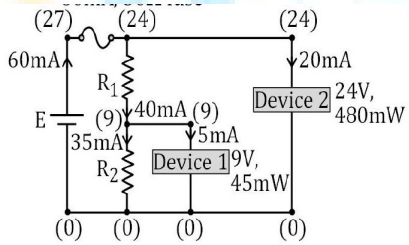
$(1) + (2) \Rightarrow 10\sqrt{\frac{3}{2}} = a\left(3\sqrt{6} + \sqrt{\frac{3}{2}}\right)$

$\frac{10}{\sqrt{2}} = a\left(3\sqrt{2} + \frac{1}{\sqrt{2}}\right)$

$10 = a(6+1)$

$a = \frac{10}{7}\ m/s^2$

32. $60\ mA, 50\Omega$ fuse



$E = 27\ V$

$24 - 9\ V = 40ma \times R_1$

$R_1 = \frac{3}{8}\ K\Omega = 375\Omega$

$9 - 0\ V = 35\ mA \times R_2$

$R_2 = \frac{9000}{35}\Omega = \frac{1800}{7}\Omega$

$\eta = \frac{480+45}{480+45+315+600+180} = \frac{525}{1620}$

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CHEMISTRY

33. Fact

$ZnS + NaCN \rightarrow Na_2[Zn(CN)_4](aq.$

NaCN is used as depressant if ZnS impurity present in PbS ore

34. For the given representation of orbital

$n = 3, \ell = 1 \Rightarrow 3p$ orbital

$\therefore$ Number of radial node $= 1$

Lorbital $= \sqrt{\ell(\ell+1)}\hbar = \sqrt{2}\hbar$

Angular part of the wave function of orbital is

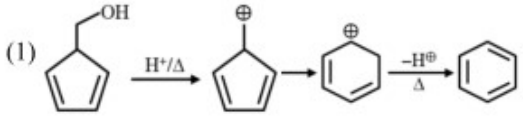
$= K\sin\theta.\sin\phi$

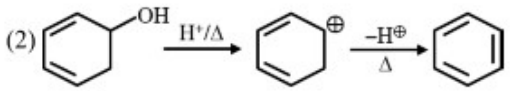
$\Rightarrow$ the orbital must be p_y

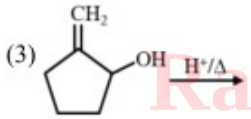
Again $3.174 = 6a^0 =$ distance of radial node

$\Rightarrow$ Probability of finding e^- is zero.

35.

(1) 

(2) 

(3) 

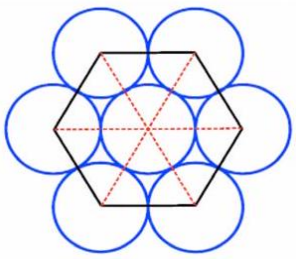
None of the formed product is aromatic

(4) $HOOC-(CH_2)_6-COOH \xrightarrow{\text{Sodalime}/\Delta}$

$\text{n-Hexane} \xrightarrow[773\ K,\ 10\text{-}20\ atm]{Cr_2O_3}$ 

36. The correct answer is option (A)

37. Packing of cells of bacteria : 2D hexagonal packing



Effective number of bacteria in hexagon $= 1 + 6 \times 1/3 = 3$

Area occupied by 3 bacteria cells = Area of hexagon

Area occupied by 1 bacteria cell $= \frac{1}{3} \times 6 \times \frac{\sqrt{3}}{4} \times (2r)^2 = 2\sqrt{3} \times 10^{-12}\ m^2$

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Area of dish $= \pi R^2 = 1.732 \times 10^{-2}\ m^2 = \sqrt{3} \times 10^{-2}\ m^2$

If N bacteria cover the total surface in mentioned manner then

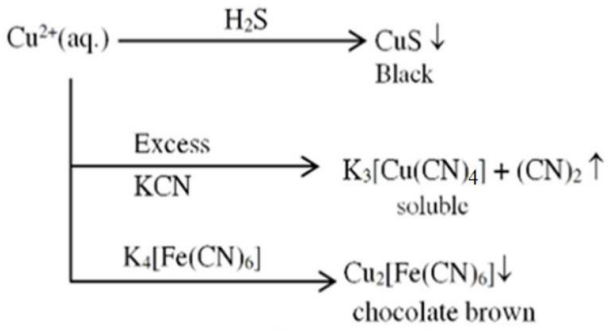
$$N = \frac{\sqrt{3} \times 10^{-2}\ m^2}{2\sqrt{3} \times 10^{-12}} = 5 \times 10^9$$

time required $(t) = \frac{N}{N_0} = (2)^{t/\tau}$ (similar to 1 st order kinetics)

$$\log \frac{N}{N_0} = \frac{t}{\tau} \log 2$$

$$t = \tau \cdot \frac{\log N / N_0}{\log 2} = 10 \times \frac{\log\left(5 \times 10^9\right)}{\log 2} = 323.33 \text{ min}$$

38.

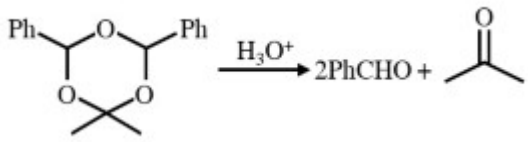


$$(CN)_2 + NaOH \rightarrow \overset{2+}{N}aCN + NaO\overset{+4}{C}N$$

CuS is soluble in aqua-regia.

$$2CuS + 8CN^- \rightarrow 2\left[Cu(CN)_4\right]^{3-} + S_2^{2-}$$

39.



(A) Compound  can give positive iodoform test

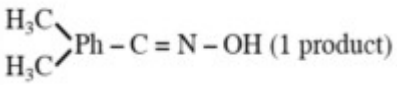
(B) PhCHO can give positive Tollens test

(C) PhCHO does not give Fehlings test

(D) $Ph - CHO + NH_2OH \rightarrow Ph - CH = N - OH$

(2 products)

$$CH_3COCH_3 + NH_2OH \rightarrow$$



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40. After long exposure to air, state of equilibrium will be achieved.

$$2Mn^{+2} + 2H_2O + O_2 \rightleftharpoons 2MnO_2(s) + 4H^+(aq.)$$

$$E_{cell} = E^0_{cell} - \frac{0.059}{4}\log\frac{[H^+]^4}{P_{O_2}\cdot[Mn^{2+}]^2}$$

$$\Rightarrow 0 = (1.23 - 1.23) - \frac{0.059}{4}\frac{[H^+]^4}{P_{O_2}\cdot[Mn^{2+}]^2}$$

$$\frac{[H^+]^4}{P_{O_2}\cdot[Mn^{2+}]^2} = 1$$

$$\Rightarrow [H^+]^4 = 0.2\times(0.8\times1\times10^{-3})^2$$

$$= 2\times64\times10^{-9}$$

$$= 2^7\times10^{-9}$$

$$pH = \frac{9}{4} - \frac{7}{4}\log2 = \frac{9-2.1}{4} = 1.725$$

41.

Species	Initial mmoles
AcOH	100
AcO^- from Barium acetate	$50\times2 = 100$
OH^- from NaOH	50

NaOH reacts with AcOH

Species m moles in resultant solution

AcOH $100 - 50 = 50$

AcO^- $100 + 50 = 150$

Resultant solution acts as acidic buffer solution

$$\therefore pH = pKa + \log\frac{[AcO^-]}{[AcOH]}$$

$$= 5 + \log\frac{150}{50} = 5 + \log3$$

$$= 5.48$$

$$pOH = 14 - 5.48 = 8.52$$

42. Cupric iodate: $Cu(IO_3)_2$

Balanced reaction 1 :

$$2Cu^{2+} + 4IO_3^- + 24I^- + 24H^+$$

$$\rightarrow 2CuI + 13I_2 + 12H_2O$$

Balanced reaction 2 :

$$2\,S_2O_3^{2-} + I_2 \rightarrow 2I^- + S_4O_6^{2-}$$

Molarity = Normality for $S_2O_3^{-2}$

$$\Rightarrow n_{Cu^{2+}} = \left(\frac{2}{13}\right)\left(\frac{1}{2}\right)n_{S_2O_3^{2-}}$$

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$= \frac{1}{13} \times 26 \times 10^{-3}$

$= 2 \times 10^{-3}$ moles

$\left[Cu^{+2}\right] = \frac{2 \times 10^{-3}}{100} \times 1000 = 2 \times 10^{-2}$ mole / lit

$Cu(IO_3)_2 \overset{K_{sp}}{\rightleftharpoons} Cu_s^{+2} + \underset{2\,s}{2IO_3^-}$

$K_{sp} = 4\,s^3 = 4 \times \left(2 \times 10^{-2}\right)^3 = 32 \times 10^{-6}$

$= 3.2 \times 10^{-5} = x \times 10^y$

$x = 3.2, y = -5$

43. $X = 5$

$Y = 2$

Alkynes for X



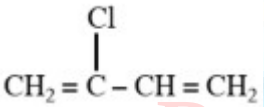
Total $= 5$

Solution for Y

2,2-Dichlorohexane and meso form of

2,5-Dichlorohexane

44. Monomer of Neoprene is chloroprene i.e, 2-chloro-1,3-butadiene



% of chlorine in chloroprene

$= \frac{35.5}{88.5} \times 100 = 40.11$

45. (i) $H_2\ S_2O_5\ (5)$

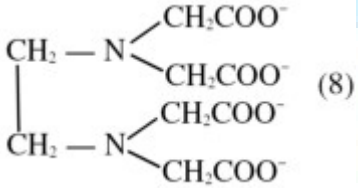
(ii) $NO_2^+NO_3^-\ (5)$

(iii) $(HPO_3)_3\ (9)$

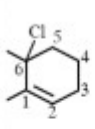
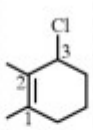
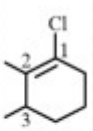
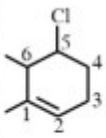
(iv) $HIO_4\ (4)$

(v) $\left[Fe(acac)_3\right]^0\ (6)$

(vi)



46.

Compounds				
Sum of locants of substituents	13	6	6	12

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47. Amylose contains $\alpha-1,4$ linkage
Amylopectin contains $\alpha-1,4$ and $\alpha-1,6$ linkages
Cellulose contains $\beta-1,4$ linkage
Lactose contains β-1,4 linkage

48. **III**

[$Fe(en)_2Cl_2]^+ : d^2sp^3$ **hybridisation**

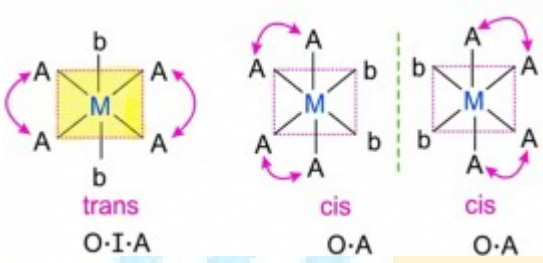
octahedral complexes of $Fe^{+3}\left(d^5\right)$ are d^2sp^3 hybrid C.A

If number of strong ligands > Number of weak ligands $\left[Cr\left(NH_3\right)_6\right]^{2+} : sp^3\ d^2$ hybridisation

Hexaammine complex of $Cr^{+2}\left(d^4\right)$ is outer orbital

complex due to low Zeff. of Cr^{+2}

$\left[Ni(en)_2\left(NH_3\right)_2\right]^{+2}$ & $\left[Fe(en)_2Cl_2\right]^{+3}$ are $M(AA)_2\ b_2$ type complexes



$\left[Ni(en)_3\right]^{2+}$ is $M(AA)_3$ type complex



All given complexes furnish cations & anions in H_2O . so, they are better electrolytic conductor

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